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Research Group: Data Analysis and Modeling in Medicine

Bachelor's Thesis in Computer Science

Data-driven camera calibration for light field camera systems

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Abstract

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1. Illustration of the pinhole camera model. Source: [3]. 2

1. Introduction

1.1. Motivation

1.2. Challenges and Problem Statement

1.3. Research Questions

The following research questions guide this thesis:

- **RQ1:** Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?
- **RQ2:** How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?
- **RQ3:** Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?

1.4. Structure of the Thesis

2. Theoretical Background

2.1. Pinhole Camera Model

Cameras are the primary devices used to capture visual information from the real world into an image. To process and analyze this information computationally, it is necessary to describe the imaging process using a camera model [2].

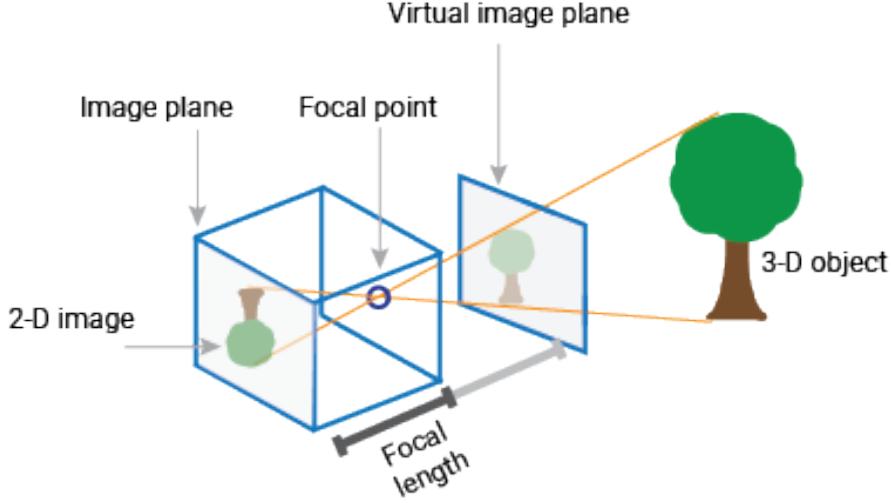


Figure 1: Illustration of the pinhole camera model. Source: [3].

The pinhole camera model describes a simple imaging system in which light rays from a 3D scene pass through a small aperture and are projected through the focal point onto a 2D image plane [2]. The aperture limits the incoming light such that each point in the scene is mapped to a single point on the image plane. In this model, the projected image is inverted due to the geometry of the projection, as shown in Figure 1.

To define the mapping from 3D world coordinates to 2D image coordinates, the model is mathematically described using two types of parameters [3].

- **Intrinsic Parameters:**

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where (f_x, f_y) are the focal lengths in pixel units along the x and y axes, s is the skew coefficient between the axes, and (c_x, c_y) is the principal point (optical center) in pixel coordinates. K is also known as the camera matrix.

- **Extrinsic Parameters:**

$$[R \mid t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

where R is a 3×3 rotation matrix representing the orientation of the camera with respect to the world coordinate system, and t is a 3×1 translation vector representing the position of the camera center in world coordinates.

These parameters form the basis of camera calibration, which aims to estimate them accurately. However in practice, cameras use lenses to focus light onto the image sensor, which introduces distortions not accounted for in the idealized pinhole model.

2.2. Lens Distortion

The camera matrix alone is not sufficient to model real imaging systems, as the ideal pinhole camera model assumes a lens-free projection. In practice, however, camera lenses introduce various optical distortions that must be taken into account to accurately describe the image formation process [3].

Lens distortion is defined as an optical aberration that causes straight lines in the real world to appear curved in the captured image [1]. Since distortion alters the geometric relationship between 3D world points and their 2D image projections, it has a direct impact on the accuracy of camera calibration. If lens distortion is not properly modeled, the estimated intrinsic and extrinsic parameters can become biased, leading to unreliable calibration results [4].

In most camera calibration models, lens distortion is represented using a small set of parametric functions. The two most common types of distortion considered in practice are radial distortion and tangential distortion.

2.2.1. Radial Distortion

- Small Definition
- figure
- Mathematics

2.2.2. Tangential Distortion

- Small Definition
- figure
- Mathematics

2.3. Camera Calibration

2.3.1. Calibration Parameters and Error Metrics

2.3.2. Target-Based Camera Calibration

2.3.3. Bundle Adjustment

2.4. Light Field Camera Systems

2.4.1. Multi-Camera Light Field Systems

2.4.2. Micro-Lens-Array-Based Light Field Cameras

2.5. State of the Art in Light Field Camera Calibration

3. Material and Method

3.1. Material

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3.1.2. Image Sets and Ground Truth Definition

3.1.3. Runtime Environment and Toolchain

3.2. Method

3.2.1. Use Case Description

3.2.2. Requirements Analysis

3.2.3. UML Class Diagram

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3.2.5. State Diagram

4. Results and Evaluation

4.1. Results of Initial Calibration

4.1.1. Intrinsic Accuracy

4.1.2. Extrinsic Accuracy

4.1.3. Reprojection and RMS Error

4.2. Results of LiFCal-Based Recalibration

4.2.1. Reprojection Error after LiFCal

4.2.2. Health Score Evolution

5. Discussion of the Results

5.1. Interpretation of Initial Calibration Results

5.2. Limitations of LiFCal in the Given Setup

5.3. Comparison between Initial Calibration and LiFCal

5.4. Comparison to state of the Art

5.5. Potential Extensions and Improvements

6. Conclusion and Outlook

6.1. Conclusion

6.2. Outlook

A. Appendix

Acknowledgment

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